

Module 24 - Variable Scope

1. What is Function Scope?

Function Scope means a variable declared inside a function can be accessed only within that function.

Example Code:

```
function outerFunction(){  
    // var is accessible throughout the function.  
    if(true){  
        var functionVar = "Im a Variable";  
    }  
    console.log(functionVar);  
}  
outerFunction();
```

2. What is Block Scope?

Block Scope means a variable declared inside a block { } can be accessed only within that block.

Example Code:

```
function blockScoped(){  
    // Accessible inside this function block.  
    let blockVariable = "Im a Block scoped variable";  
    if(true){  
        // Accessible only inside this if block.  
        const blockVariable1 = "Im a const variable";  
        console.log(blockVariable);  
        console.log(blockVariable1);  
    }  
}  
blockScoped();
```

3. What is Global Scope?

Global Scope means a variable declared outside all functions and blocks can be accessed from anywhere in the program.

Example Code:

```
var a = 10;

let b = 20;

const c = 30;

function accessGlobalLocalVar(){

    // Local variables can be declared here.

    // var a = 101;

    // let b = 202;

    // const c = 303;

    function innerFunction(){

        // Inner function can access variables from outer/global scope.

        // var a = 100;

        // let b = 200;

        // const c = 300;

        console.log(a + b + c);

    }

    innerFunction();

    console.log(a + b + c);

}

accessGlobalLocalVar();

console.log(a + b + c);
```

- ❑ **var** → Function scoped, can redeclare and reassign.
- ❑ **let** → Block scoped, can reassign but cannot redeclare.
- ❑ **const** → Block scoped, cannot reassign or redeclare.